

# Chengzu Li

Webpage <https://chengzu-li.github.io/> [Google Scholar] [Semantic Scholar]  
Contact +44-07783-497657  
cl917@cam.ac.uk

## Education

2023 - Now **University of Cambridge**  
PhD student supervised by Prof. Anna Korhonen, Dr. Ivan Vulić and Prof. Serge Belongie.  
Fully funded by Cambridge Trust.  
Visiting student at Belongie Lab (2025.8 - 2025.11)

2022 - 2023 **University of Cambridge**  
M.Phil. in Advanced Computer Science (Distinction 81.7/100)  
Supervised by Prof. Simone Teufel. Fully funded by Cambridge Trust.

2018 - 2022 **Xi'an Jiaotong University**  
Bachelor of Engineering in Automation (91.96/100, top 5%). Minor in Fintech.

2021 **University of Hong Kong**  
Exchange student in Computer Engineering (4.08/4.3)

## Publications

### Conferences

(\* indicates equal contribution)

2026 **Visual Planning: Let's Think Only with Images**  
Yi Xu\*, Chengzu Li\*, Han Zhou\*, Xingchen Wan, Caiqi Zhang, Anna Korhonen, Ivan Vulić  
**ICLR Oral** [Github]  
- Featured at **X(Twitter)**, **Medium** and **etc.(CHN)**  
- Community influence with over 290 Github stars

**Deforming Videos to Masks: Flow Matching for Referring Video Segmentation**  
Zanyi Wang, Dengyang Jiang, Liuzhuozheng Li, Sizhe Dang, Chengzu Li, Harry Yang, Guang Dai, Mengmeng Wang, Jingdong Wang  
**ICLR** [Github]

2025 **Information Loss in Vision-Language Models**  
Wenyan Li, Raphael Tang, Chengzu Li, Caiqi Zhang, Ivan Vulić, Anders Søgaard  
**EMNLP Findings** [Github]

**Enriching Patent Claim Generation with European Patent Dataset**  
Lekang Jiang, Chengzu Li, Stefan Goetz  
**EMNLP Findings**

**Imagine While Reasoning in Space: Multimodal Visualization-of-Thought**  
Chengzu Li\*, Wenshan Wu\*, Huanyu Zhang, Yan Xia, Shaoguang Mao, Li Dong, Ivan Vulić, Furu Wei  
**ICML** [Github]  
- Featured at **IEEE Spectrum**, **TWIML AI Podcast** and **etc.(CHN)**

- Community influence over 60 Github stars

### **Large Language Models are Miscalibrated In-Context Learners**

Chengzu Li, Han Zhou, Goran Glavaš, Anna Korhonen, Ivan Vulić

**ACL Findings** [Github]

2024

### **TOPVIEWRS: Vision-Language Models as Top-View Spatial Reasoners**

Chengzu Li\*, Caiqi Zhang\*, Han Zhou, Nigel Collier, Anna Korhonen, Ivan Vulić

**EMNLP Oral** [Github]

- Featured at **TWIML AI Podcast**

- Non-archival at ACL 2023 SpLU Robo Workshop (**Oral**), ECCV 2024 Eval-Fomo Workshop

### **Semantic Map-based Generation of Navigation Instructions**

Chengzu Li, Chao Zhang, Simone Teufel, Rama Sanand Doddipatla, Svetlana Stoyanchev

**COLING**

2023

### **Generating Data for Symbolic Language with Large Language Models**

Jiacheng Ye, Chengzu Li, Lingpeng Kong, Tao Yu

**EMNLP** [Github]

### **Binding Language Models in Symbolic Languages**

Zhoujun Cheng\*, Tianbao Xie\*, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, Tao Yu

**ICLR Spotlight** [Github]

2022

### **UnifiedSKG: Unifying and Multi-Tasking Structured Knowledge Grounding with Text-to-Text Language Models**

Tianbao Xie\*, Chen Henry Wu\*, Peng Shi, Ruiqi Zhong, Torsten Scholak, Michihiro Yasunaga, Chien-Sheng Wu, Ming Zhong, Pengcheng Yin, Sida I. Wang, Victor Zhong, Bailin Wang, Chengzu Li, Connor Boyle, Ansong Ni, Ziyu Yao, Dragomir Radev, Caiming Xiong, Lingpeng Kong, Rui Zhang, Noah A. Smith, Luke Zettlemoyer, Tao Yu

**EMNLP Oral** [Github]

*Pre-prints*

### **How Well Do Models Follow Visual Instructions? VIBE: A Systematic Benchmark for Visual Instruction-Driven Image Editing**

Huanyu Zhang\*, Xuehai Bai\*, Chengzu Li\*, Chen Liang, Haochen Tian, Haodong Li, Ruichuan An, Yifan Zhang, Anna Korhonen, Zhang Zhang, Liang Wang, Tieniu Tan

### **Thinking in Frames: How Visual Context and Test-Time Scaling Empower Video Reasoning**

Chengzu Li\*, Zanyi Wang\*, Jiaang Li\*, Yi Xu, Han Zhou, Huanyu Zhang, Ruichuan An, Dengyang Jiang, Zhaochong An, Ivan Vulić, Serge Belongie, Anna Korhonen

### **Latent Sketchpad: Autoregressive Visual Latent Generation for Interpretable Visual Thoughts in MLLMs**

Huanyu Zhang\*, Wenshan Wu\*, Chengzu Li, Ning Shang, Yan Xia, Yangyu Huang, Yifan Zhang, Li Dong, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

### **11PLUS-BENCH: Demystifying Multimodal LLM Spatial Reasoning with Cognitive-Inspired Analysis**

Chengzu Li, Wenshan Wu, Huanyu Zhang, Qingtao Li, Zeyu Gao, Yan Xia, José Hernández-Orallo, Ivan Vulić, Furu Wei

**Scaling and Beyond: Advancing Spatial Reasoning in MLLMs Requires New Recipes**

Huanyu Zhang\*, Chengzu Li\*, Wenshan Wu, Shaoguang Mao, Yifan Zhang, Haochen Tian, Ivan Vulić, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

**Reinforcement Learning for Better Verbalized Confidence in Long-Form Generation**

Caiqi Zhang\*, Xiaochen Zhu\*, Chengzu Li, Nigel Collier, Andreas Vlachos

**Professional Experiences**

|      |                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>GenAI Group, Microsoft Research</b><br>Research Intern                                                                        |
| 2023 | <b>Toshiba Cambridge Research Laboratory</b><br>Research Intern                                                                  |
| 2021 | <b>Shanghai Artificial Intelligence Lab</b><br>Research Intern<br><b>HKU CS Research Internship Programme</b><br>Research Intern |

**Academic Service**

**Paper Reviewing**

|     |                      |
|-----|----------------------|
| NLP | ACL ARR 2024, 2025   |
| AI  | ICLR 2026, ICML 2026 |
| CV  | CVPR 2026            |

**Selected Awards & Grants**

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | One of three nominations by University of Cambridge for the Apple AI/ML PhD Fellowship<br>Postgraduate Research Fund from Jesus College, Cambridge<br>Lambda Research Grant (\$2000 USD in GPU credits) |
| 2024 | Postgraduate Research Fund from Jesus College, Cambridge<br>Scholar of Jesus College, Cambridge (for outstanding academic performance)                                                                  |
| 2023 | PhD Scholarship from Cambridge Trust (Fully-Funded)                                                                                                                                                     |
| 2022 | Masters Scholarship from Cambridge Trust (Fully-Funded)                                                                                                                                                 |
| 2019 | National Scholarship (Top 1%, highest honor for undergraduates in China)                                                                                                                                |

**Media Coverage & Presentations**

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>Invited Talk at HKUST (GZ):</b> Reason with Multimodal Minds in Space<br><b>JiQiZhiXin (CHN):</b> Thinking with Images Alone: RL Fosters a New Paradigm for Reasoning Models<br><b>XinZhiYuan (CHN):</b> Beyond CoT! Directly Visualize the Multimodal Reasoning Process<br><b>TWIML AI Podcast with Sam Charrington:</b> Multimodal Visualization-of-Thought<br><b>IEEE Spectrum:</b> “Thinking” Visually Boosts AI Problem Solving |
| 2024 | <b>BMVA:</b> Trustworthy Multimodal Learning with Foundation Models                                                                                                                                                                                                                                                                                                                                                                     |